

Assignment 5: Data Visualization

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Visualization

Directions

1. Rename this file `<FirstLast>_A05_DataVisualization.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

H.P

Set up your session

1. Set up your session. Load the tidyverse, here & cowplot packages, and verify your home directory. Read in the NTL-LTER processed data files for nutrients and chemistry/physics for Peter and Paul Lakes (use the tidy NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv version in the Processed_KEY folder) and the processed data file for the Niwot Ridge litter dataset (use the NEON_NIWO_Litter_mass_trap_Processed.csv version, again from the Processed_KEY folder).
2. Make sure R is reading dates as date format; if not change the format to date.

```
#1
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.1
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(here)
```

```
## here() starts at /home/guest/EDE_Spring2026
```

```
library(cowplot)
```

```
##  
## Attaching package: 'cowplot'  
##  
## The following object is masked from 'package:lubridate':  
##  
##     stamp
```

```
getwd()
```

```
## [1] "/home/guest/EDE_Spring2026"
```

```
processed_data = here("Data/Processed_KEY")  
PeterPaul.chem.nutrients <- read_csv(  
  here(processed_data, "NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv")  
)
```

```
## Rows: 23008 Columns: 15  
## -- Column specification -----  
## Delimiter: ","  
## chr   (1): lakename  
## dbl  (13): year4, daynum, month, depth, temperature_C, dissolvedOxygen, irra...  
## date  (1): sampleddate  
##  
## i Use 'spec()' to retrieve the full column specification for this data.  
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
NiwotRidge.litter <- read_csv(  
  here(processed_data, "NEON_NIWO_Litter_mass_trap_Processed.csv")  
)
```

```
## Rows: 1692 Columns: 13  
## -- Column specification -----  
## Delimiter: ","  
## chr   (7): plotID, trapID, functionalGroup, qaDryMass, nlcdClass, plotType, g...  
## dbl   (5): dryMass, subplotID, decimalLatitude, decimalLongitude, elevation  
## date  (1): collectDate  
##  
## i Use 'spec()' to retrieve the full column specification for this data.  
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
#2  
# Make sure reading dates as Date format  
class(PeterPaul.chem.nutrients$sampleddate)
```

```
## [1] "Date"
```

```
class(NiwotRidge.litter$collectDate)
```

```
## [1] "Date"
```

Define your theme

3. Build a theme and set it as your default theme. Customize the look of at least two of the following:

- Plot background
- Plot title
- Axis labels
- Axis ticks/gridlines
- Legend

```
#3
mytheme <- theme_classic(base_size = 14) +
  theme(plot.background = element_rect(color = "white"),
        plot.title = element_text(color = "black", size = 16, face = "bold"),
        axis.title = element_text(color = "black", size = 12),
        panel.grid = element_line(color = "gray"),
        legend.position = "right"
  )
theme_set(mytheme)
```

Create graphs

For numbers 4-7, create ggplot graphs and adjust aesthetics to follow best practices for data visualization. Ensure your theme, color palettes, axes, and additional aesthetics are edited accordingly.

4. [NTL-LTER] Plot total phosphorus (tp_ug) by phosphate (po4), with separate aesthetics for Peter and Paul lakes. Add line(s) of best fit using the `lm` method. Adjust your axes to hide extreme values (hint: change the limits using `xlim()` and/or `ylim()`).

```
#4
total_phosphorus <-
  ggplot(PeterPaul.chem.nutrients,
    aes(x = po4,
        y = tp_ug,
        color = lakename,
        shape = lakename)) +
  geom_point(size = 2) +
  geom_smooth(method = "lm", linewidth = 1) +
  labs(
    title = "Total Phosphorus by Phosphate",
    x = "Phosphate (µg/L)",
    y = "Total Phosphorus (µg/L)",
    color = "Lake",
    shape = "Lake"
  ) +
  coord_cartesian(xlim = c(0, 20),
```

```

ylim = c(0, 60))

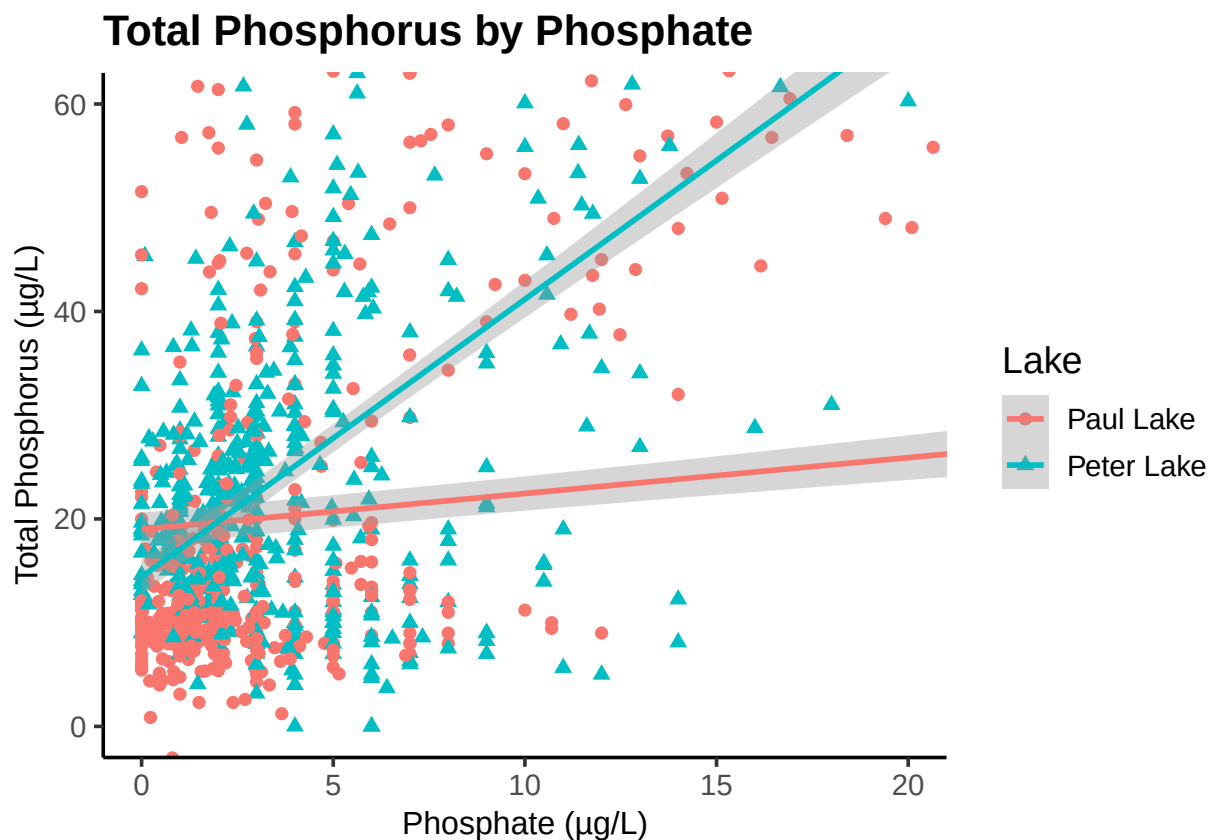
print(total_phosphorus)

```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 21946 rows containing non-finite outside the scale range
## ('stat_smooth()').
```

```
## Warning: Removed 21946 rows containing missing values or values outside the scale range
## ('geom_point()').
```



5. [NTL-LTER] Make three separate boxplots of (a) temperature, (b) TP, and (c) TN, with month as the x axis and lake as a color aesthetic. Then, create a cowplot that combines the three graphs. Make sure that only one legend is present and that graph axes are aligned. Show all months, even those where no data was collected.

Tips: * Recall the discussion on factors in the lab section as it may be helpful here. * Setting an axis text in your theme to `element_blank()` removes the axis text (useful when multiple, aligned plots use the same axis values) * Setting a legend's position to "none" will remove the legend from a plot. * Individual plots can have different relative sizes when combined using `cowplot`.

```

#5
PP <- PeterPaul.chem.nutrients %>%
  mutate(month = factor(month, levels = 1:12, labels = month.abb)) %>%
  filter(!is.na(month))

temp <-
  ggplot(PP,
    aes(x = month,
        y = temperature_C,
        fill = lakename)) +
  geom_boxplot() +
  scale_x_discrete(drop = FALSE) +
  labs(x = NULL, y = "Temperature", fill = "Lake", title = "Temperature by Month") +
  theme(axis.text.x = element_blank(),
        legend.position = "right")

TP <-
  ggplot(PP,
    aes(x = month,
        y = tp_ug,
        fill = lakename)) +
  geom_boxplot() +
  scale_x_discrete(drop = FALSE) +
  labs(x = NULL, y = "TP (µg/L)", fill = "Lake", title = "TP by Month") +
  theme(axis.text.x = element_blank(),
        legend.position = "none")

TN <-
  ggplot(PP,
    aes(x = month,
        y = tn_ug,
        fill = lakename)) +
  geom_boxplot() +
  scale_x_discrete(drop = FALSE) +
  labs(x = NULL, y = "TN (µg/L)", fill = "Lake", title = "TN by Month") +
  theme(legend.position = "none")

plot_grid(temp, TP, TN, nrow = 3, align = "v", rel_heights = c(0.8, 1, 1))

```

```

## Warning: Removed 3566 rows containing non-finite outside the scale range
## ('stat_boxplot()').

```

```

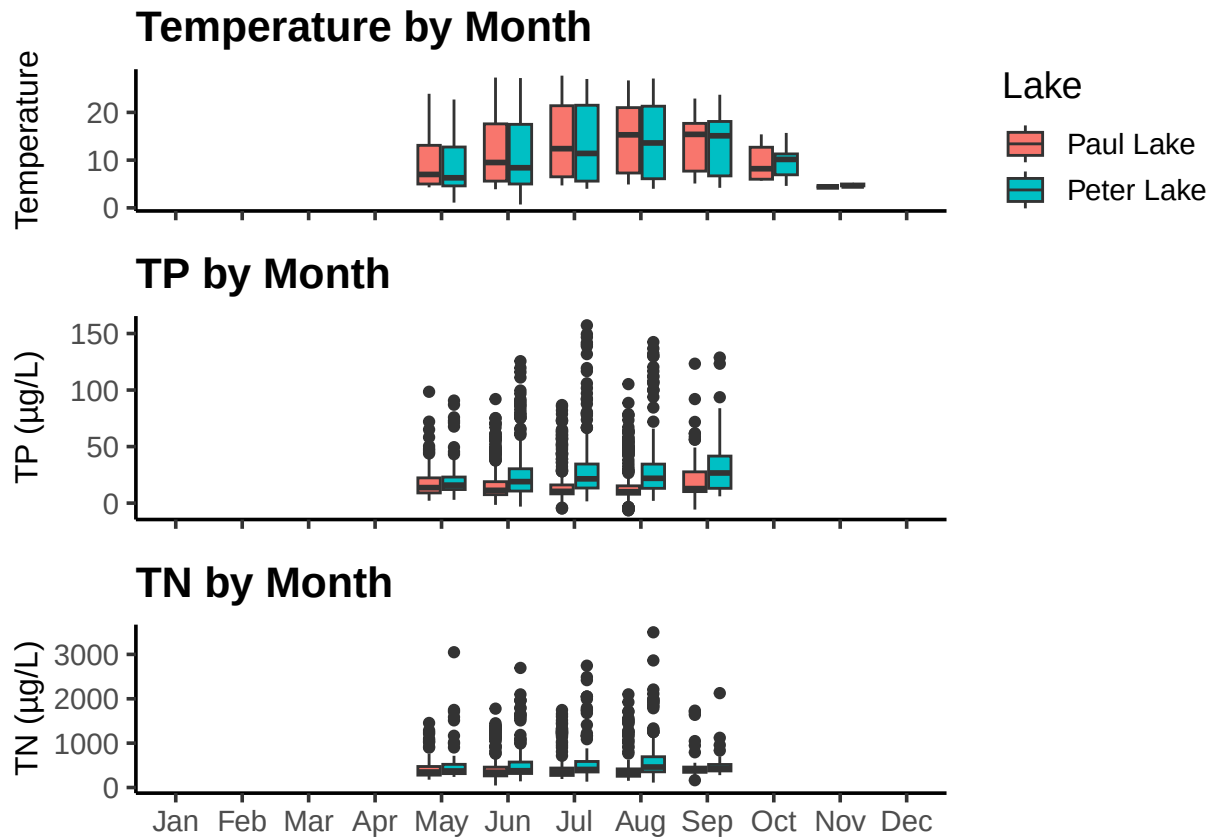
## Warning: Removed 20729 rows containing non-finite outside the scale range
## ('stat_boxplot()').

```

```

## Warning: Removed 21583 rows containing non-finite outside the scale range
## ('stat_boxplot()').

```



Question: What do you observe about the variables of interest over seasons and between lakes?

Answer: Both TP and TN concentration increase during the summer (typically from June to August), along with the increasing temperature. Both TP and TN show the strong variability and lots of extreme values in summer. Also, Peter Lake tends to have higher concentration of both TP and TN than Paul Lake, indicating a more nutrient-rich pattern.

- [Niwot Ridge] Plot a subset of the litter dataset by displaying only the “Needles” functional group. Plot the dry mass of needle litter by date and separate by NLCD class with a color aesthetic. (no need to adjust the name of each land use)
- [Niwot Ridge] Now, plot the same plot but with NLCD classes separated into three facets rather than separated by color.

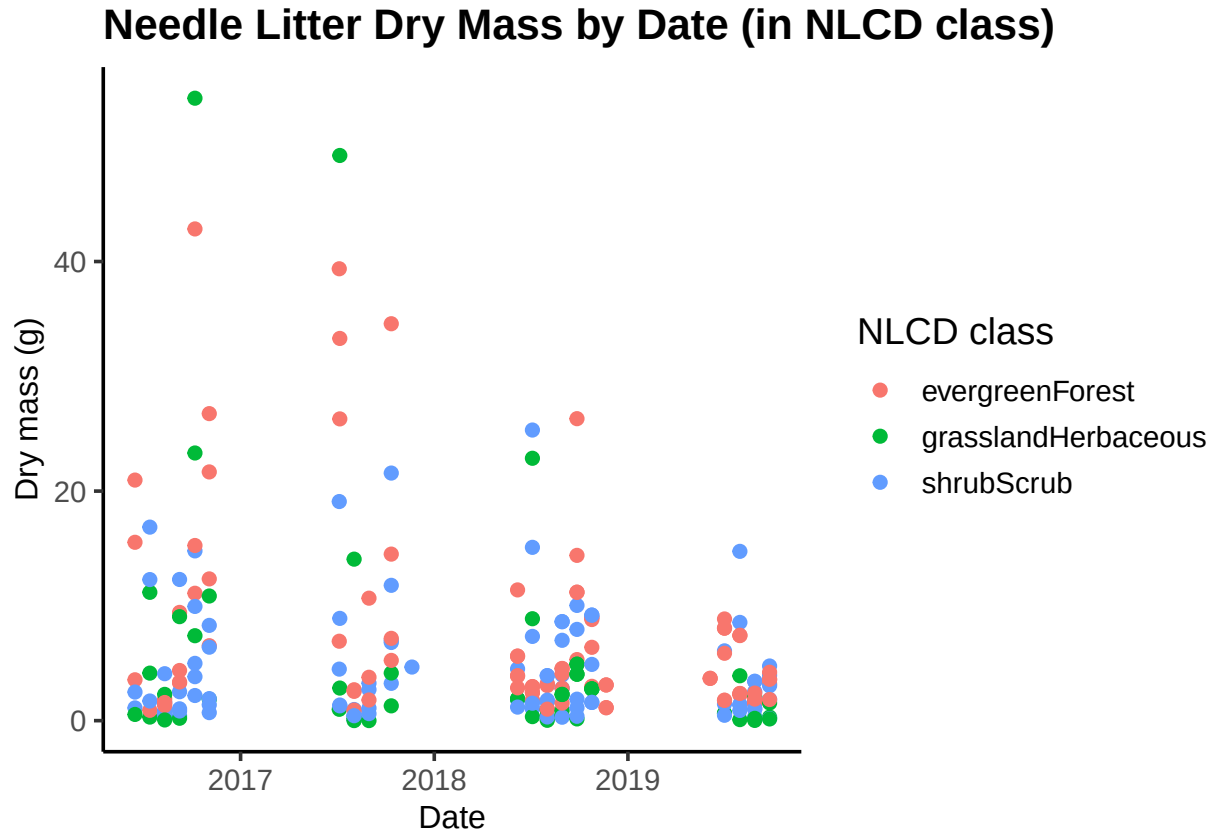
```
#6
needles <- NiwotRidge.litter %>%
  filter(functionalGroup == "Needles")

Q6_dry_mass <-
  ggplot(needles,
    aes(x = collectDate,
        y = dryMass,
        color = nlcdClass)) +
  geom_point(size = 2) +
  labs(
    title = "Needle Litter Dry Mass by Date (in NLCD class)",
```

```

x = "Date",
y = "Dry mass (g)",
color = "NLCD class")
print(Q6_dry_mass)

```

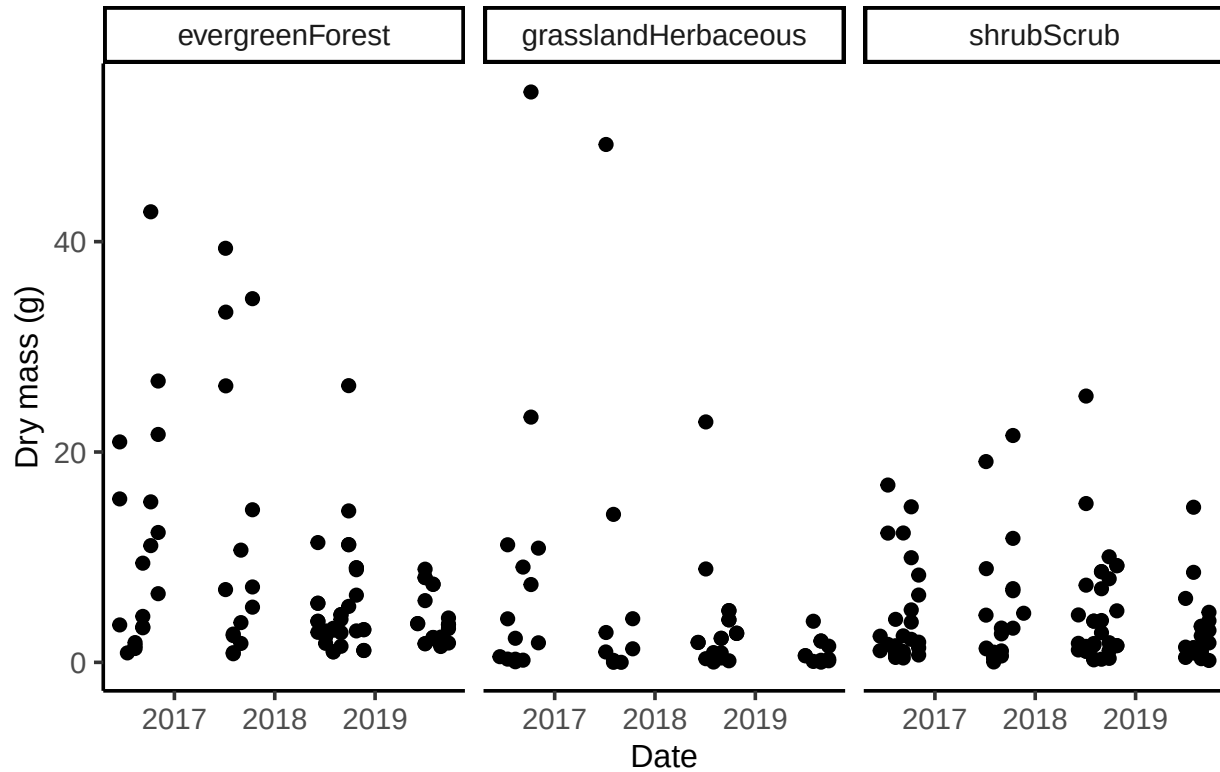


```

#7
Q7_dry_mass <-
  ggplot(needles,
    aes(x = collectDate,
        y = dryMass)) +
  geom_point(size = 2) +
  facet_wrap(~nlcdClass, ncol = 3) +
  labs(
    title = "Needle Litter Dry Mass by Date (in NLCD class)",
    x = "Date",
    y = "Dry mass (g)")
print(Q7_dry_mass)

```

Needle Litter Dry Mass by Date (in NLCD class)



Question: Which of these plots (6 vs. 7) do you think is more effective, and why?

Answer: I think the Plot 7 is more effective because it indicates the variation of needle litter dry mass by date clearly. Also, by separating each land cover class into its own panel, this facet plot (Plot 7) allows the comparison of magnitude across classes to be more straightforward. However, the Plot 6 has too many colored and overlapping points, which makes it difficult to observe the dry mass patterns for each NLCD class separately.